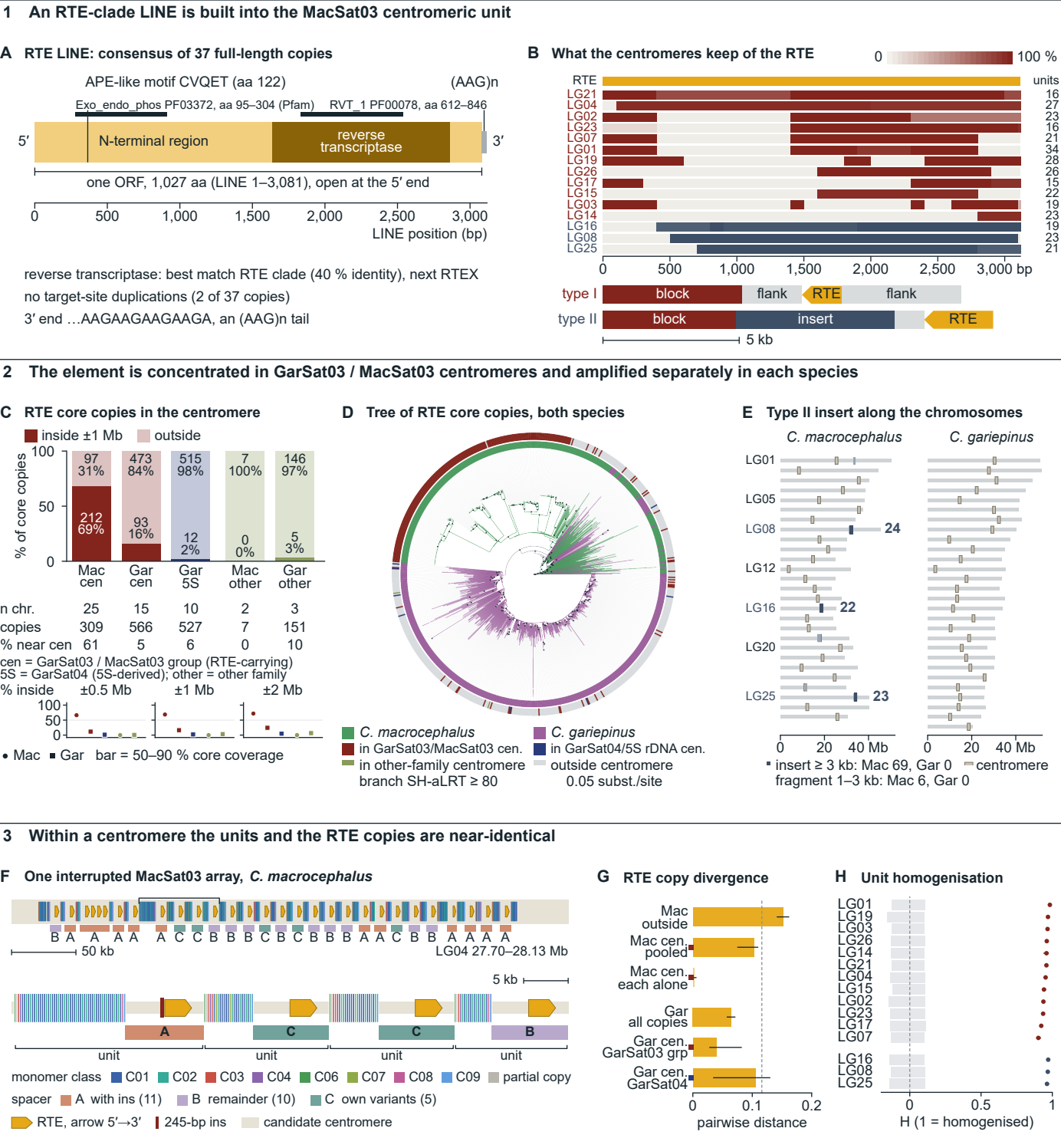

1 An RTE-clade LINE is built into the MacSat03 centromeric unit
A RTE LINE: consensus of 37 full-length copies
APE-like motif CVQET (aa 122)
(AAG)n
Exo_endo_phos PF03372, aa 95–304 (Pfam)
RVT_1 PF00078, aa 612–846
5′
N-terminal region
reverse transcriptase
3′
one ORF, 1,027 aa (LINE 1–3,081), open at the 5′ end
0
500
1,000
1,500
2,000
2,500
3,000
LINE position (bp)
reverse transcriptase: best match RTE clade (40 % identity), next RTEX
no target-site duplications (2 of 37 copies)
3′ end …AAGAAGAAGAAGA, an (AAG)n tail
B What the centromeres keep of the RTE
0
100 %
RTE
units
LG21 16
LG04 27
LG02 23
LG23 16
LG07 21
LG01 34
LG19 28
LG26 26
LG17 15
LG15 22
LG03 19
LG14 23
LG16 19
LG08 23
LG25 21
0
500
1,000
1,500
2,000
2,500
3,000 bp
type I
block
flank
RTE
flank
type II
block
insert
RTE
5 kb
2 The element is concentrated in GarSat03 / MacSat03 centromeres and amplified separately in each species
C RTE core copies in the centromere
inside ±1 Mb
outside
% of core copies
100
50
0
97
31%
212
69%
473
84%
93
16%
515
98%
12
2%
7
100%
0
0%
146
97%
5
3%
Mac cen
Gar cen
Gar 5S
Mac other
Gar other
n chr. 25 15 10 2 3
copies 309 566 527 7 151
% near cen 61 5 6 0 10
cen = GarSat03 / MacSat03 group (RTE-carrying)
5S = GarSat04 (5S-derived); other = other family
% inside ±0.5 Mb ±1 Mb ±2 Mb
100
50
0
• Mac ▪ Gar bar = 50–90 % core coverage
D Tree of RTE core copies, both species
C. macrocephalus
in GarSat03/MacSat03 cen.
in other-family centromere
branch SH-aLRT ≥ 80
C. gariepinus
in GarSat04/5S rDNA cen.
outside centromere
0.05 subst./site
E Type II insert along the chromosomes
C. macrocephalus
C. gariepinus
LG01
LG05
LG08
24
LG12
LG16
22
LG20
LG25
23
0
20
40 Mb
insert ≥ 3 kb: Mac 69, Gar 0
fragment 1–3 kb: Mac 6, Gar 0
centromere
3 Within a centromere the units and the RTE copies are near-identical
F One interrupted MacSat03 array, C. macrocephalus
B A A A A A C C B B B C B C B B B A A C B B A A A A
50 kb
LG04 27.70–28.13 Mb
5 kb
A
C
C
B
unit
unit
unit
unit
monomer class C01 C02 C03 C04 C06 C07 C08 C09 partial copy
spacer A with ins (11) B remainder (10) C own variants (5)
RTE, arrow 5′→3′
245-bp ins
candidate centromere
G RTE copy divergence
Mac outside
Mac cen. pooled
Mac cen. each alone
Gar all copies
Gar cen. GarSat03 grp
Gar cen. GarSat04
0
0.1
0.2
pairwise distance
H Unit homogenisation
LG01
LG19
LG03
LG26
LG14
LG21
LG04
LG15
LG02
LG23
LG17
LG07
LG16
LG08
LG25
0
0.5
1
H (1 = homogenised)